

In T 49/99 the board ruled that information modelling was a non-technical intellectual activity, but that the purposive use of information modelling in the context of a solution to a technical problem could contribute to the technical character of an invention.

In T 858/02 the board held that an electronic message was not automatically excluded from patentability as a presentation of information. It depended on whether the message was defined by its structure or its content. A computer data structure was not under all circumstances excluded from patentability; the fact that in the present case the instructions defined a structure of the message did not automatically lead to the conclusion that a "format (i.e. data structure)" was unpatentable.

In T 1351/04 an index file used for the purpose of controlling the computer "along the path leading to the desired data" was considered to contribute to the solution of a technical problem. See also decisions T 1902/10, T 2539/12, T 2330/13.

9.2.11 Assessment of features relating to mathematical algorithms

In T 1784/06 the board stated that as the algorithm is a mathematical (inter alia Boolean) method and mathematical methods as such are deemed to be non-inventions, mathematical algorithms may contribute to the technical character of an invention only in so far as they serve a technical purpose (Art. 52(2) and (3) EPC), (see also T 208/84; T 1227/05, OJ 2007, 574; T 1358/09; T 306/10; T 566/11; T 2035/11; T 2249/13; T 2330/13).

In T 208/84 the board made a distinction between abstract concepts on the one hand and technical processes involving and modifying a "physical entity", such as an electrical signal, on the other hand. It held that a basic difference between a mathematical method and a technical process could be seen in the fact that a mathematical method or a mathematical algorithm was carried out on numbers (whatever these numbers may represent) and provided a result also in numerical form, the mathematical method or algorithm being only an abstract concept prescribing how to operate on the numbers. No direct technical result was produced by the method as such. In contrast thereto, if a mathematical method was used in a technical process, that process was carried out on a physical entity (which could be a material object but equally an image stored as an electric signal) by some technical means implementing the method and provided as its result a certain change in that entity.

In T 1161/04 the invention related to an apparatus for rebalancing a stock index. The board held that apart from the input means, processing means and output means, the features of the claim did not contribute to the technical character of the invention. They merely defined the computer operations necessary to implement an algorithm for rebalancing a capitalization-weighted stock index. The board further concluded that the data input to the claimed apparatus was cognitive and had no technical function. The processing performed on them comprised classification, scaling and redistribution. These steps concerned exclusively the cognitive content of the data (their numerical value) which was pure information processing and as such excluded as a mental act.

In T.1814/07 the application related to an automated collection and analysis patient care system and method for ordering and prioritising multiple health disorders to identify an index disorder. The board held that if a mathematical method is used in a technical process, and this process is carried out on a physical entity by some technical means implementing the method and provides as its result a change in that entity, it contributes to the technical character of the invention as a whole.

In T.2418/12 an algorithm suggested related terms. This was considered not to be a technical problem; whether terms were "related" to each other was a cognitive or linguistic matter and not a technical issue (T.1358/09, T.2230/10, T.2439/11). This was in line with the case law of the boards of appeal, which generally holds that **algorithmic efficiency** is not a technical effect (cf. T.1784/06, T.42/10, T.1370/11).

a) Design

In T.471/05 the main request was directed to a method of designing an optical system, the method consisting essentially of designing the optical system so that substantially all light rays imaged by the optical system between two predetermined points on the optical axis of the system satisfied the algebraic condition specified in the claim. The board held that the claim merely formulated a series of mathematical and optical abstract concepts without properly requiring a physical, technical implementation. Neither the claimed design method nor the resulting "design" required a technical activity or a technical entity. It followed that the subject-matter was the "mere design" of an optical system and encompassed purely abstract and conceptual implementations. More particularly, the claimed method could be carried out as a purely mental act or as a purely mathematical design algorithm.

In T.835/10 the board held that "de-automating" or undoing (in a computer-implemented method) the automation performed by a prior art software – could not in general be considered to be inventive. In particular, the board could not see an inventive activity in leaving the optimisation task mainly to the designer and providing him with the necessary aid to perform that task (reporting the evaluation parameters for the current design and providing him with a GUI for modifying the design).

b) Simulations

In T.1227/05 (OJ 2007, 574) the board came to the conclusion that the claimed numerical simulation of a noise-affected circuit described by a model featuring input channels, noise input channels and output channels and a system of differential or algebraic differential equations was a functional technical feature. This simulation constituted an adequately defined technical purpose for a computer-implemented method functionally limited to that purpose. The board held that specific technical applications of computer-implemented simulation methods were themselves to be regarded as modern technical methods which formed an essential part of the fabrication process and preceded actual production, mostly as an intermediate step. In that light, such simulation methods could not be denied a technical effect merely on the ground that they did not yet incorporate the physical end product.

In G.1/19 the invention related to a method of testing – by simulation – a modelled environment with respect to pedestrian crowd movement. The Enlarged Board answered the questions of law referred to it as follows:

A computer-implemented simulation of a technical system or process that is claimed as such can, for the purpose of assessing inventive step, solve a technical problem by producing a technical effect going beyond the simulations implementation on a computer.

For that assessment it is not a sufficient condition that the simulation is based, in whole or in part, on technical principles underlying the simulated system or process.

The answers to the first and second questions are no different if the computer-implemented simulation is claimed as part of a design process, in particular for verifying a design.

When considering the existing case law on simulations, the Enlarged Board agreed with the findings of T.1227/05 and T.625/11 if they were understood as being that the claimed simulation processes in those particular cases possessed an intrinsically technical function. It also did not see a need to require a direct link with (external) physical reality in every case. However, it held that there were rather strict limits for the consideration of potential or merely calculated technical effects according to the COMVIK approach (T.641/00). The approach of T.1227/05, that the simulation constituted an adequately defined technical purpose for a numerical simulation method if it was functionally limited to that purpose should not be taken as a generally applicable criterion of the COMVIK approach for computer-implemented simulations, since the findings of T.1227/05 were based on specific circumstances which did not apply in general. In the Enlarged Board's opinion, the COMVIK approach was suitable for the assessment of computer-implemented simulations. Like any other computer-implemented inventions, numerical simulations may be patentable if an inventive step can be based on features contributing to the technical character of the claimed simulation method. In the opinion of the Enlarged Board, when the COMVIK approach is applied to simulations, the underlying models form boundaries, which may be technical or non-technical. In terms of the simulation itself, these boundaries were not technical. However, they may contribute to technicality if, for example, they were a reason for adapting the computer or its functioning, or if they formed the basis for a further technical use of the outcomes of the simulation (e.g. a use having an impact on physical reality). In order to avoid patent protection being granted to non-patentable subject-matter, such further use had to be at least implicitly specified in the claim. The same applied to any adaptations of the computer or its functioning. The same considerations applied to simulations claimed as part of a design process. A design process was normally a cognitive exercise. However, the Enlarged Board found that it certainly could not be ruled out that in future cases there may be steps within a design process involving simulations which did contribute to the technical character of the invention.

On question one, the Enlarged Board concluded that no group of computer-implemented inventions could be a priori excluded from patent protection. The COMVIK approach required an assessment of the technical contribution of the individual features of computer-

implemented inventions. Like any other computer-implemented method, a simulation without an output having a direct link with physical reality could still solve a technical problem (see also T 489/14 of 26 November 2021).

On question two the Enlarged Board concluded that it had been established in the COMVIK approach that, depending on the technical context, features that were non-technical per se could still contribute to the technical character of a claimed invention, just as features that were technical per se would not necessarily contribute to it. In view of this, the Enlarged Board was of the opinion that it was neither a sufficient nor a necessary condition that a numerical simulation was based, at least in part, on technical principles that underlay the simulated system or process.

On question three, the Enlarged Board did not see any need for the application of special rules if a simulation was claimed as part of a design process.

At issue in T 625/11, was a method to establish at least one threshold value of a parameter for operating a nuclear reactor with a view to making better use of the reactor's capacity. The board found that establishing a threshold value for the first operating parameter lent the claim a technical dimension going beyond the mere interaction of the numerical-simulation algorithm and the computer system, the established parameter having a purpose closely linked to operating a nuclear reactor, regardless of whether or not it was actually used for that. The board thus acknowledged that the approach taken in T 1227/05 (OJ 2007, 574) was appropriate.

c) Database management systems and information retrieval systems

In T 107/87 the board held that a data coding rule for identifying and eliminating statistical redundancy contributed to the solution of a technical problem where it was used to reduce the amount of data to be stored or transmitted. This meant that if a computer-implemented method included steps of losslessly compressing and decompressing intermediate results to reduce the amount of memory space required for storing those results, at least those steps would make a technical contribution. The implementation of the coding rule would normally still be algorithmic in nature (see also T 650/13).

In T 1242/04 (OJ 2007, 421) the invention related to a system for providing product-specific data in a service station. Since the claims featured a central database for recording the required status and an archive store for recording the actual status which communicated with each other by computing means, the board found that they had technical character.

In T 279/05 the invention related to determining airline seat availability. The invention involved a mixture of technical aspects, e.g. servers, and non-technical aspects, e.g. airline seat availability and yield management. Database querying was considered to be a technical field by the board.

In T 1924/17 the board saw no reason why relational database management systems should be non-technical, if it was accepted that database management systems in general

were technical. The board noted that its position was not that all features implemented in (relational) database management systems contributed by virtue of this fact alone and independent of their nature to the technical character of an invention. The board reviewed decisions relating to accessing data base management systems (T 1242/04 (OJ EPO 2007, 421), T 279/05, T 862/05, T 658/06, T 1500/08, T 963/09, T 104/12, T 1965/11) and information retrieval (T 1569/05, T 1316/09, T 309/10, T 2230/10, T 598/14). In the light of these decisions, the board in T 1924/17 summarised that structured declarative queries, which are used for retrieving data managed in a relational database management system, normally have precise, formally defined semantics and the database management system then retrieves the specified data set as a result. It explained that relational database management systems typically executed such queries by determining an efficient query execution plan based on cost estimates for the necessary internal operations of the computer system. Such database management systems were software platforms for the centralised control of data. Features of these platforms often had a technical character, as they had been designed based on engineering considerations concerning the efficient exploitation of the computer system as a technical system. Information retrieval systems typically had to formally calculate a semantic similarity of documents, which is typically regarded as involving non-technical considerations and being based on subjective criteria and the content (semantics) of the documents to be retrieved.

d) Navigation systems

In T 2035/11 the application mainly related to navigation systems that could be tailored to a user's particular wishes. The board held that optimisation algorithms are mathematical methods. It noted that **mathematical algorithms may contribute to the technical character** of an invention only in so far as they serve a **technical purpose** (see e.g. decision T 1784/06). The purpose of the algorithm was the mere display of an optimal path to the user for cognitive processing. The user could act on the information, but did not need to. As stated in decision T 1670/07, a technical effect may arise from either the provision of data about a technical process, regardless of the presence of the user or its subsequent use, or from the provision of data (including data that on its own is excluded, e.g. produced by means of an algorithm) that is applied directly in a technical process. In the case at issue, the data was produced by means of an algorithm and was not applied directly in a technical process, so that neither possibility applied. The German Federal Supreme Court had come to a similar conclusion in respect of a navigation system offering the user the possibility to exclude road segments based on a user-selectable property such as the road segment being a toll road (see BGH, 18 December 2012, X ZR 3/12 – Routenplanung).

In T 651/12 the board stated that what were meant in Art. 52(2)(a) EPC were merely abstract mathematical methods, i.e. calculations for the sake of the calculation. However, in the case at issue, which was essentially the technical implementation in a map display apparatus of the method for displaying a bird's eye view map, the outcome of the calculation was used for a technical purpose, namely to **display information in an ergonomically improved manner**. The board further stated that also the calculation as such in that case had in the board's judgment clear technical aspects: the technical effect